

The Power of Vaccination From Smallpox Eradication to COVID-19 in Record Time

A Data-Driven Overview of Two Centuries of Vaccine Impact (1796–2020)

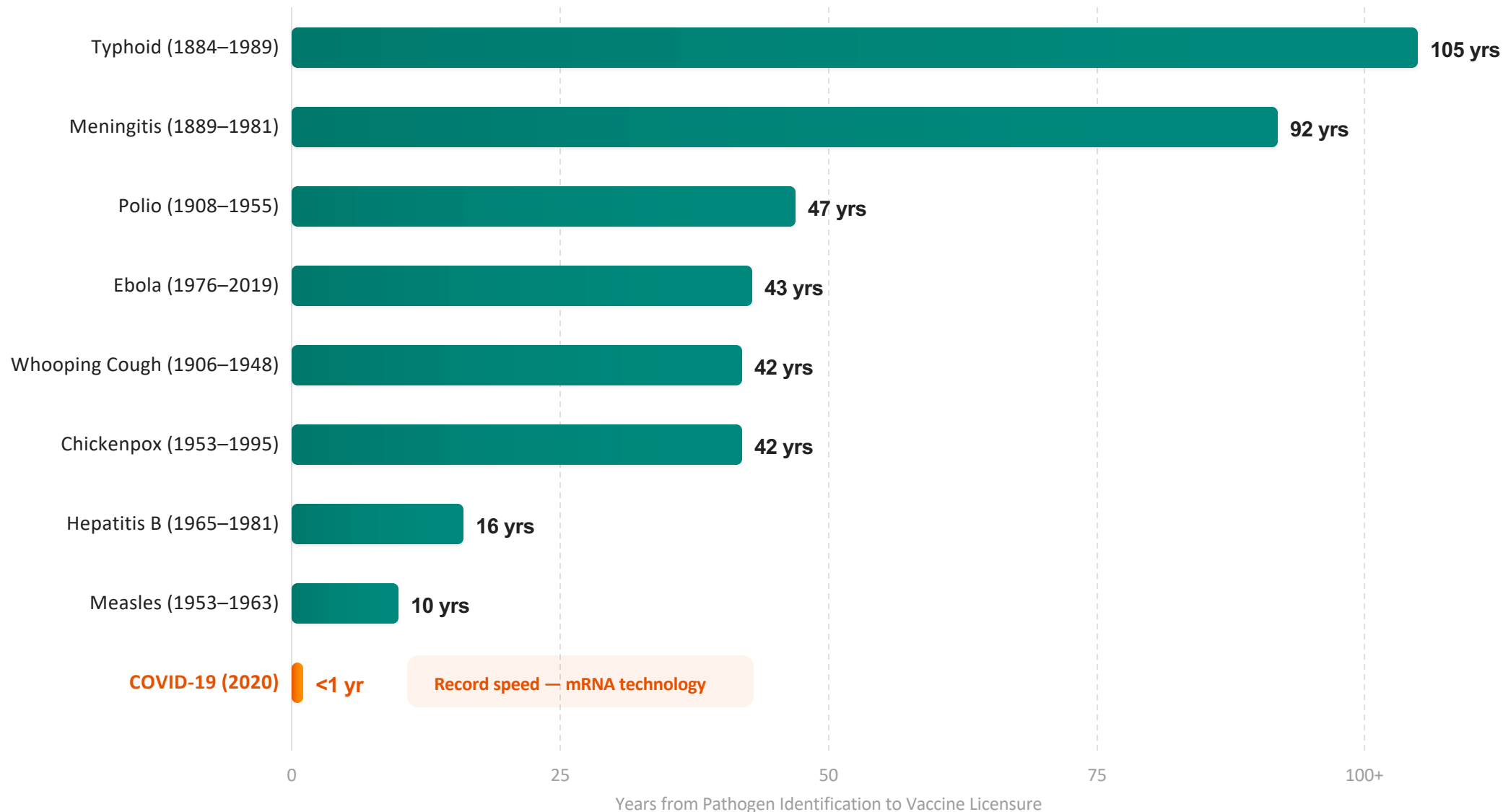
Source: Our World in Data

For Public Health Professionals & Policy Analysts



From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Modern tools — genome sequencing and mRNA technology — have compressed vaccine development from over a century to under one year.



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Vaccines have driven 100% elimination of multiple deadly diseases; remaining gaps highlight where investment is still needed.

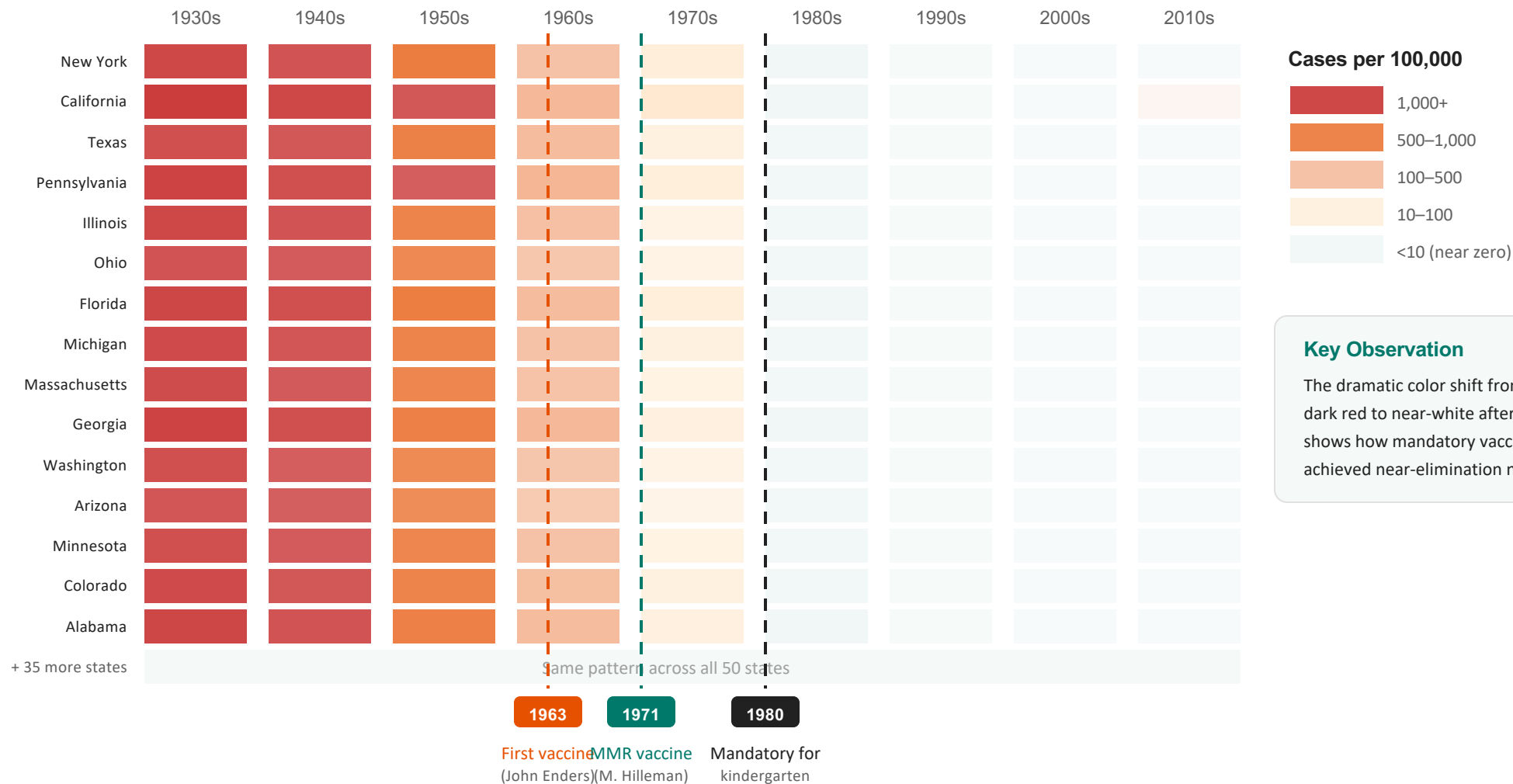
Disease	Pre-Vaccine cases/M/yr	Post-Vaccine cases/M/yr	Case Reduction	Pre-Vaccine deaths/M/yr	Death Reduction	
Diphtheria	158	0	100%	13.7	100%	✓
Smallpox	250	0	100%	2.9	100%	✓
Acute Polio	141	0	100%	10	100%	✓
Measles	3,044	0.2	99.99%	2.5	100%	
Rubella	242	0.04	99.98%	0.09	100%	
Pertussis	1,534	52	96.6%	30.8	99.7%	
Hepatitis A	465	51	89%	0.5	88.7%	
Acute Hepatitis B	273	44	83.9%	1	83.6%	
Pneumococcal	233	139	40.5%	24	31.3%	⚠

✓ 100% elimination (cases & deaths)

⚠ Significant remaining burden

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Before 1963, nearly every state saw 100–1,000+ cases per 100,000. After the 1980 mandate, cases dropped to near zero nationwide.



Smallpox: The Only Human Disease Eradicated Through Vaccination

The WHO-led campaign eliminated smallpox in just 18 years — a unique success enabled by the virus infecting only humans.



Why Was Smallpox Uniquely Suited for Eradication?



Human-Only Host

The virus infected only humans
— no animal reservoir



Distinguishable Symptoms

Cases could be identified
and isolated reliably



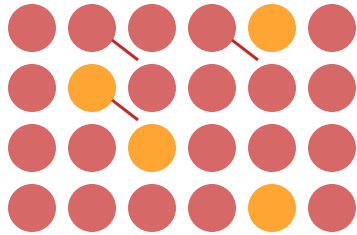
Cheap, Effective Vaccine

Vaccine was affordable and
highly effective worldwide

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

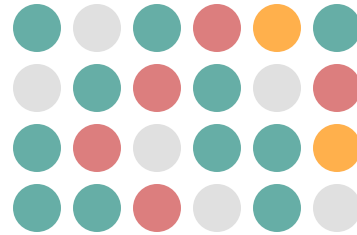
When enough people are vaccinated, pathogens cannot spread effectively — shielding newborns, immunocompromised, and cancer patients.

No Vaccination



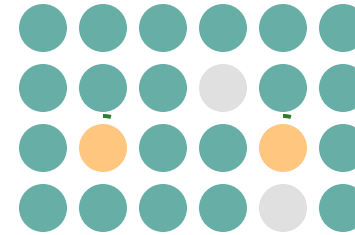
Pathogen spreads freely
Vulnerable individuals infected

Partial Vaccination



Some spread still occurs
Vulnerable still at risk

Herd Immunity Achieved



Spread is blocked
Vulnerable are protected

 Vaccinated  Unvaccinated (healthy)  Infected  Vulnerable (cannot be vaccinated)  Protected

Who Cannot Be Vaccinated — and Relies on Herd Immunity?

Newborns

Too young to receive most vaccines.
Rely entirely on surrounding vaccinated population for protection against disease.

Immunocompromised

Immune systems too weak to mount vaccine response. Includes organ transplant recipients and those with autoimmune conditions.

Cancer Patients

Chemotherapy suppresses the immune system, making vaccination unsafe or ineffective during treatment.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

While vaccines have eliminated multiple diseases, significant burden remains for pneumococcal, hepatitis A, and hepatitis B.

Incomplete Case Reduction

Pneumococcal Disease



Hepatitis A



Acute Hepatitis B



For comparison: Diphtheria, Smallpox, Polio = 100% elimination

Key Barriers to Full Coverage



Global Access Gaps

Not everyone has access to vaccines globally. Unequal distribution leaves populations at risk.



Supply Shortages

Vaccine supply shortages put children at risk, especially in low- and middle-income countries.



Trust & Understanding

Poor scientific understanding reduces vaccine trust and uptake in many communities.



Priority: Improved access, newer vaccines, and community engagement are needed to close the remaining gaps.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done



100% Elimination Achieved

Diphtheria, smallpox, and polio have been completely eliminated in the United States.



Development Timelines Collapsed

From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.



Measles Near-Zero Nationwide

Cases dropped to near zero across all 50 US states after the 1980 kindergarten vaccination mandate.



Only Disease Eradicated

Smallpox remains the only human disease eradicated through vaccination, certified in 1980.



Coverage Gaps Persist

Pneumococcal (40.5%), hepatitis A (89%), and hepatitis B (83.9%) still have significant remaining burden.



Call to Action for Global Health Systems

1. Expand global vaccine access — ensure equitable distribution to all populations
2. Invest in next-generation vaccines — close gaps for pneumococcal disease and hepatitis
3. Build public trust — improve scientific understanding and community engagement